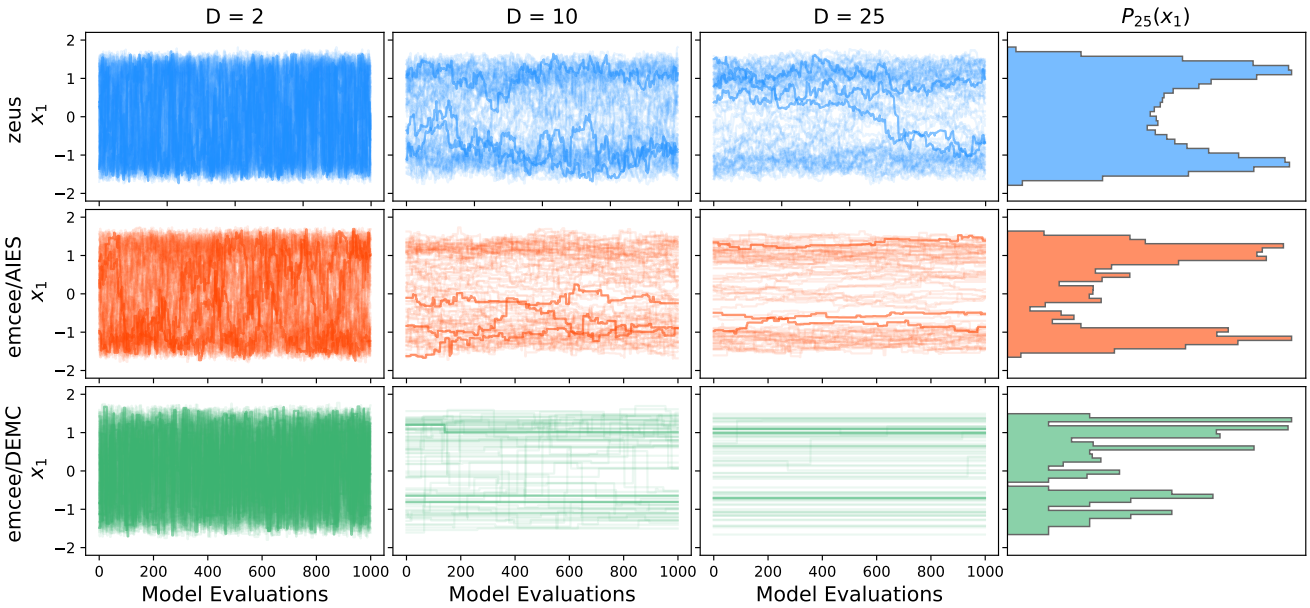

Figure 8. The figure shows numerical results (i.e. walker trajectories/chains for the first parameter) demonstrating the performance of the three ensemble MCMC methods in the case of the ring target distribution in 2, 10 and 25 dimensions respectively. The last column illustrates the 1-D marginal posterior corresponding to the first parameter x_1 estimated directly from the samples for the 25-dimensional case. One can notice here that in 10 and 25 dimensions both `emcee` methods mix very slowly. In the 25-dimensional case almost all of `emcee`/DEMC's walkers are unable to move and the autocorrelation time is effectively infinite.

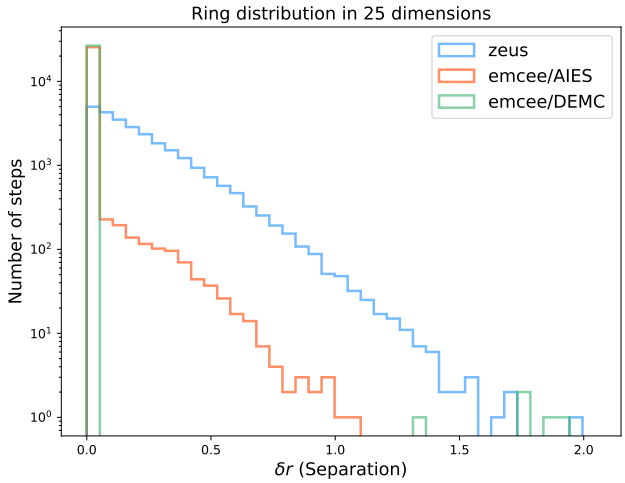

Figure 9. This figure shows the distribution of step sizes of walkers for the three different samplers in the case of a ring target distribution in $D = 25$. It is important to note here that both `emcee` algorithms exhibit a peak at zero separation; `zeus` on the other hand does not. The existence of the zero-peak in `emcee` is due to the high number of rejected proposals (i.e. low acceptance rate).

6 in which the computational cost until convergence (i.e. number of model evaluations) for a single walker diminishes and then reaches a plateau as the number of walkers is increased. We notice however that, at the level of a single walker, the computational cost of `emcee`/AIES is significantly higher compared to that of either `zeus` or `emcee`/DEMC. This is due to the way that different samplers choose the directions along which walkers move. In particular, both `zeus` and `emcee`/DEMC define a direction vector as the difference between two walkers from the complementary ensemble, thus two walkers are required to define a direction. On the other hand, `emcee`/AIES requires only a single walker from the complementary ensemble as the direction is defined by the difference between the updated walker and the complementary one. This stark contrast between the way those samplers choose their direction vectors lies at the heart of the difference in the computational cost of `emcee`/AIES as compared to `zeus` and `emcee`/DEMC in the limit of low number of walkers. In order to dive a little deeper into this, we can compute the exact number of possible directions for all three methods. Since `emcee`/AIES requires only a single walker from the complementary ensemble the number of available directions is equal to the size of the complementary ensemble. On the other hand, `zeus`'s and `emcee`/DEMC's requirement for a pair of walkers means that the number of available directions is equal to $\binom{n}{2}$, meaning the 2–combination from a set of n walkers that comprise the complementary ensemble. Clearly, as shown in Figure 7, the latter increases faster with the size of the complementary ensemble, thus explaining the larger variety of possible directions available in the case of `zeus` and `emcee`/DEMC compared to `emcee`/AIES.

The discussion so far was about the computational cost of convergence in terms of the number of model evaluations for a single walker. Of course, the ensemble of walkers consists by definition of more than a single walker. Therefore, in order to compute the total number of model evaluations required until the ensemble converges we need to multiply the results of the single walker with the total number of walkers. Those results are presented in the right panel of Figure 6. From this plot we can see that both `zeus` and `emcee`/DEMC converge faster when the number of walkers is close to its minimum value i.e. $2 \times D$. `emcee`/AIES on the other hand prefers a higher number of walkers (i.e. $32 \times D$) in order to overcome the sparsity of available directions in the limit of low number of walkers. This, however, means that even if we choose the optimal number of walkers for `emcee`/AIES it would still converge slower than either `zeus` or `emcee`/DEMC. Furthermore, we cannot know *a priori* the optimal number of walkers for `emcee`/AIES unlike for `zeus` and `emcee`/DEMC in which the optimal size of the ensemble is close to $2 \times D$. Finally, the faster convergence of `zeus` compared to

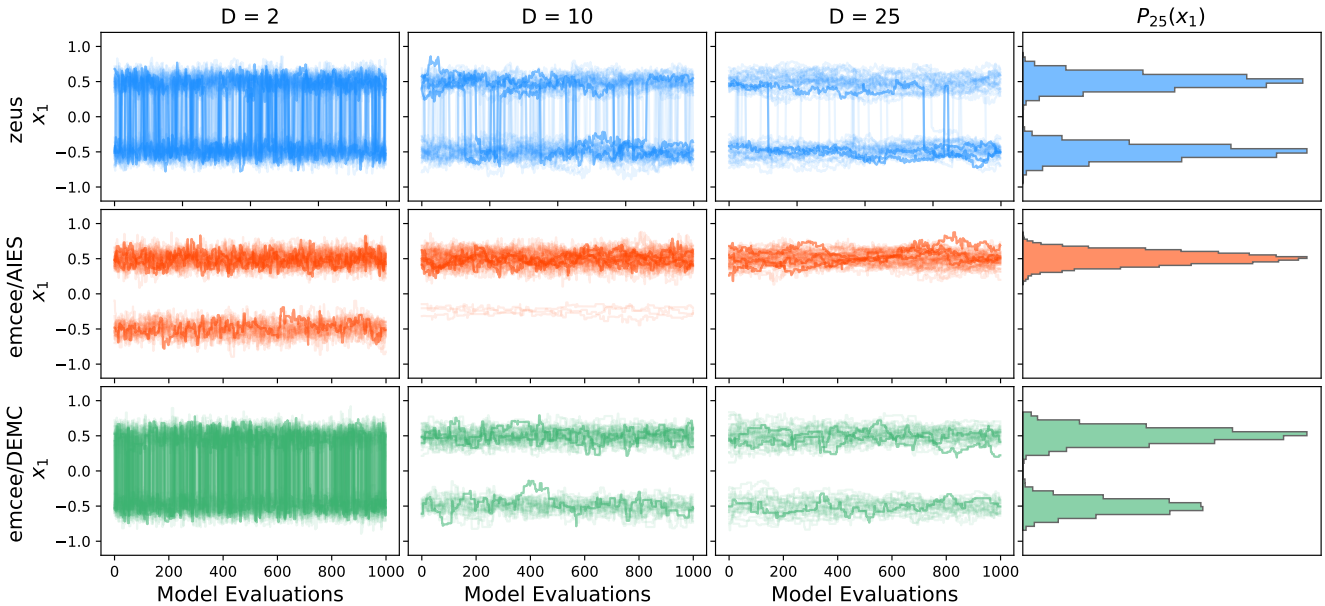

Figure 10. The figure shows numerical results (i.e. walker trajectories/chains for the first parameter) demonstrating the performance of the three ensemble MCMC methods in the case of a two-component Gaussian mixture target distribution in 2, 10 and 25 dimensions respectively. The last column illustrates the 1-D marginal posterior corresponding to the first parameter x_1 estimated directly from the samples for the 25-dimensional case. Whereas all three samplers make valid within-mode proposals, it is only `zeus` that manages to perform between-mode jumps and thus sample correctly from the target distribution in the 10 and 25-dimensional cases. Between-mode jumps are paramount in order to distribute the probability mass correctly between different modes.

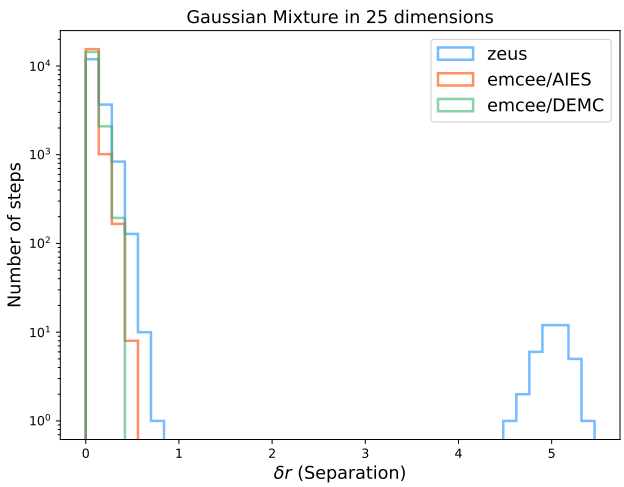

Figure 11. This figure shows the distribution of step sizes of walkers for the three different samplers in the case of a two-component Gaussian mixture target distribution in $D = 25$. It is important to note here that both `emcee` algorithms exhibit a peak at zero separation; `zeus` on the other hand does not due to its non-rejection basis.

`emcee`/DEMC can be attributed to the local adaptation that the former performs by extending the length of the slice and thus allowing larger steps in parameter space.

3.1.2 *The ring distribution*

The ring distribution defined as

$$\ln P(x) = -\left[\frac{(x_n^2 + x_1^2 - a)^2}{b}\right]^2 - \sum_{i=1}^{n-1}\left[\frac{(x_i^2 + x_{i+1}^2 - a)^2}{b}\right]^2, \quad (3)$$

where $a = 2$, $b = 1$ and n is the total number of parameters; this is an artificial target distribution that exhibits strong non-linear correlations between its parameters. This aspect of the ring distribution allows us to demonstrate the locally adaptive nature of `zeus`. Whereas `emcee`/AIES and `emcee`/DEMC use a single global proposal scale for all regions of the parameter space, `zeus` has the ability to adjust its proposal scale locally by expanding the slice appropriately. As expected, this will allow `zeus` to sample efficiently even in cases in which strong non-linear correlations are present. Looking at Figure 8 one can see that `zeus` manages to generate multiple samples efficiently even in high dimensions. On the other hand, `emcee`/AIES and `emcee`/DEMC do not efficiently produce valid proposals: for `emcee`/AIES this leads to an inefficient random walk, characterised by small steps; for `emcee`/DEMC the acceptance rate almost vanishes beyond $D = 2$. The expected squared jump distance of each method for the case of $D = 25$ is shown in Table 1. It is important to note here that out of the three samplers only `zeus` manages to converge in all three cases (i.e. in 2, 10 and 25 dimensions). `emcee`/AIES and `emcee`/DEMC on the other hand converge successfully only in 2 dimensions.

To explain this result one only has to look at the distribution of walker steps of the different methods at Figure 9. `zeus`'s steps extend to large distances in parameter space whereas most of `emcee`/AIES's and `emcee`/DEMC's steps are rejected (i.e. shown as zero in the histogram). We can see that `emcee`/DEMC manages to perform some long distance steps but those are few and there is almost nothing in between. It is clear from this and the previous toy examples that the $\gamma = 2.38/\sqrt{D}$ scaling of `emcee`/DEMC's scale factor does not generalise well beyond the Gaussian case.

3.1.3 *The two-component Gaussian mixture distribution*

One other important aspect of astronomical posterior distributions is the fact that many of them exhibit multiple peaks. Multimodality can

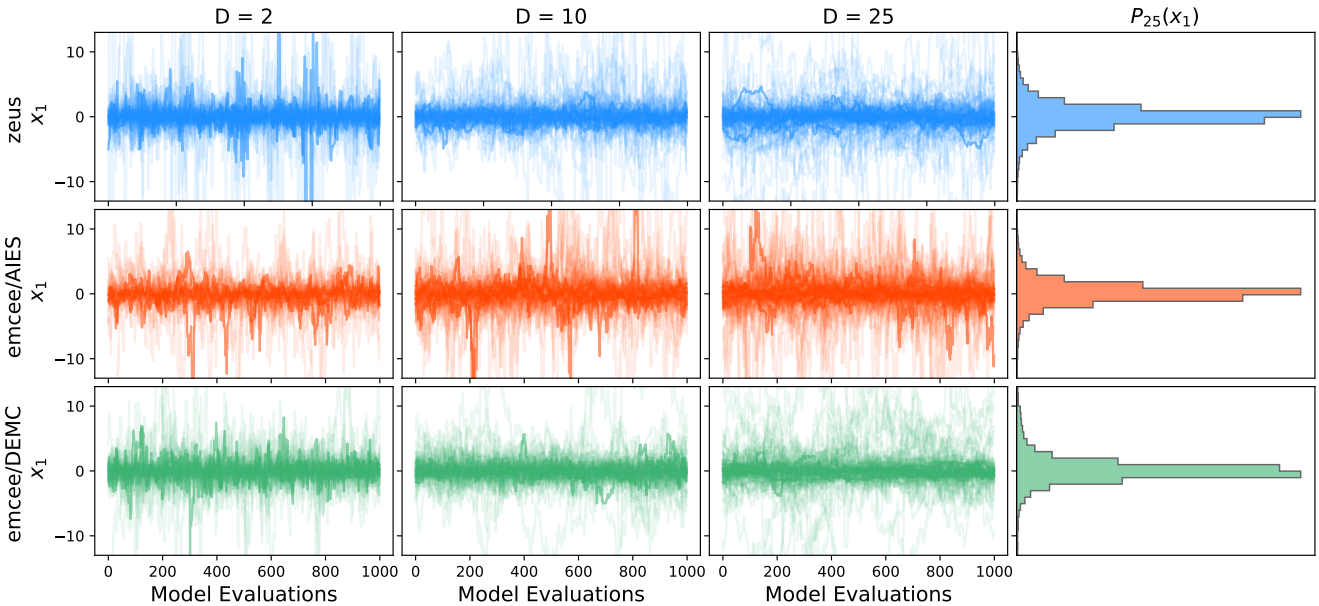

Figure 12. The figure shows numerical results (i.e. walker trajectories/chains for the first parameter) demonstrating the performance of the three ensemble MCMC methods in the case of the Student's t-distribution with 2 degrees of freedom in 2, 10 and 25 dimensions respectively. The last column illustrates the 1-D marginal posterior corresponding to the first parameter x_1 estimated directly from the samples for the 25-dimensional case.

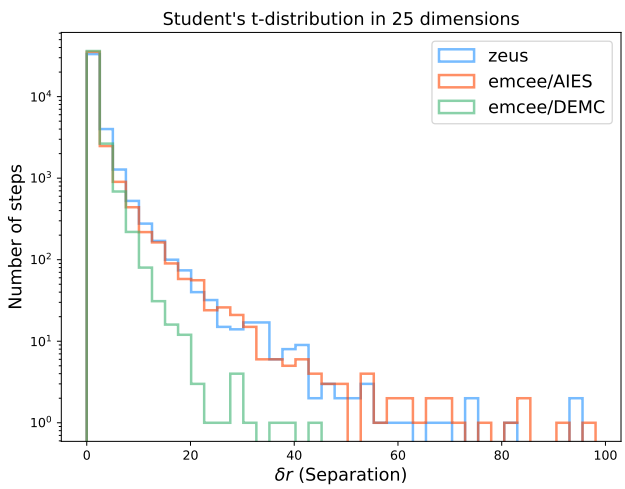

Figure 13. This figure shows the distribution of step sizes of walkers for the three different samplers in the case of the Student's t-distribution with 2 degrees of freedom in $D = 25$. `zeus` and `emcee`/AIES exhibit similar distributions whereas `emcee`/DEMC performs shorter steps.

arise either from non-linear models or sparse and uninformative data. In either case, multimodal target distributions present a formidable challenge for most MCMC methods. Perhaps the simplest example of such a distribution is the two-component Gaussian mixture. In this example we will position the two, equal-mass, components at $-\mathbf{0.5}$ and $+\mathbf{0.5}$ respectively with standard deviation of 0.1. Sampling from multimodal distributions requires two types of proposals, local proposals that sample different modes individually and global proposals that transfer walkers from one mode to the other. For this reason we will make use of `zeus`'s `GlobalMove` that uses a Dirichlet Process Gaussian Mixture model of the ensemble to efficiently propose between-mode and within-mode steps.

As seen in Figure 10, `zeus`'s walkers manage to move from one mode to the other frequently enough for mixing to be efficient even in the $D = 25$ case. Out of `emcee`/AIES and `emcee`/DEMC, only the latter proposes valid steps from one mode to the other in the $D = 2$ case. As for the $D = 25$ case, one can see in Figure 11 that `zeus`'s walkers perform numerous jumps whereas `emcee`'s walkers are unable to do so. The ability of the walkers to jump from mode to mode is of paramount importance if we want to sample correctly from the target distribution. Lack of such proposals will lead to an improper probability mass ratio between the two modes and thus biased inference. The expected squared jump distance of each method for the case of $D = 25$ is shown in Table 1.

Clustering-based proposals have also been applied to MH-type ensemble MCMC methods but as shown in Karamanis & Beutler (2020), they fail to generate valid proposals in problems with moderate number of dimensions. The reason is, as discussed in Section 3, that MH has to propose a valid point in the other mode. In other words, whereas Ensemble Slice Sampling only needs to determine the direction of the other mode relative to the chosen walker correctly, MH needs to guess both the direction and the distance, a task that rapidly becomes very hard as the number of dimensions rises.

3.1.4 The Student's t-distribution

The fourth toy example tests the case in which the target distribution is characterised by heavy-tails. In order to demonstrate `zeus`'s ability to sample efficiency is such cases we chose to use the multivariate Student's t-distribution with 2 degrees of freedom. The aforementioned density exhibits heavier tails than a normal distribution which means that it is more likely to produce samples that are far away from the mean. The t-distribution arises when estimating the mean of a normally distributed sample with unknown standard deviation and small size. The probability density function of a p–dimensional Student's t-distribution with ν degrees of freedom is given by:

$$P(x) = \frac{\Gamma[(\nu+p)/2]}{\Gamma(\nu/2)\nu^{p/2}\pi^{p/2}|\boldsymbol{\Sigma}|^{1/2}} \exp\left[1+\frac{1}{\nu}(\mathbf{x}-\boldsymbol{\mu})^T\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right]^{-\frac{(\nu+p)}{2}}, \tag{4}$$